

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET D — ASSERTION & REASONING SPECIAL****CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-D-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET D - Assertion & Reasoning Special):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Concepts & Theoretical Principles.
5. **Section B (Q11 – Q18):** Evidentiary Proofs, Case Inferences & Historical/Geographical Mechanisms.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Institutional/Analytical Complexities.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10) [10 MARKS]**Q.1**

Assertion (A): Almost all convective weather phenomena such as clouds, rain, and storms are restricted to the troposphere.

Reason (R): The troposphere contains approximately 75% of the total atmospheric mass and virtually all of its atmospheric water vapour.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. **(D)** Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): Commercial jet airliners frequently prefer cruising in the lower stratosphere.

Reason (R): The lower stratosphere is relatively free from vertical convective turbulence and severe tropospheric storm disturbances.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): Temperature decreases systematically with increasing altitude throughout every layer of the Earth's atmosphere.

Reason (R): In the stratosphere, temperature increases with altitude due to the absorption of solar ultraviolet radiation by ozone molecules.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): The normal environmental lapse rate in the troposphere averages a decrease of approximately 6.5°C per 1,000 metres of elevation.

Reason (R): The atmosphere is heated from below primarily by longwave terrestrial radiation emitted by the Earth's surface rather than direct solar insolation.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): The Coriolis force deflects moving air masses to their right in the Northern Hemisphere and to their left in the Southern Hemisphere.

Reason (R): The Coriolis effect is an apparent inertial deflection generated by the Earth's rotation on its axis from west to east.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): The Coriolis force attains its maximum magnitude at the equator and drops to zero at the geographic poles.

Reason (R): The magnitude of the Coriolis force is directly proportional to the sine of the latitude ($2\Omega v \sin \phi$).

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.7

Assertion (A): Equatorial regions experience a permanent low-pressure trough known as the Intertropical Convergence Zone (ITCZ).

Reason (R): Intense solar insolation year-round heats the equatorial surface, causing air to expand, become buoyant, and ascend convectively.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.8

Assertion (A): Subtropical high-pressure belts located around 30°N and 30°S are thermally induced by extreme cold surface temperatures.

Reason (R): Subtropical high-pressure ridges are dynamically induced by the descending limb of the Hadley cell circulation where upper-tropospheric air subsides.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.9

Assertion (A): Naturally occurring greenhouse gases are vital for sustaining complex life on planet Earth.

Reason (R): Without the natural greenhouse effect, Earth's mean surface temperature would drop to approximately -18°C, freezing global hydrospheric systems.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.10

Assertion (A): Nitrogen and oxygen gases are the primary greenhouse gases responsible for absorbing infrared heat in the lower atmosphere.

Reason (R): Diatomic homonuclear molecules like N_2 and O_2 cannot absorb infrared wavelengths because they lack changing dipole moments during vibration.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

SECTION B: EVIDENTIARY PROOFS & MECHANISMS (Q.11 TO Q.18) [8 MARKS]**Q.11**

Assertion (A): The Southwest Monsoon brings torrential summer rains to the Indian subcontinent.

Reason (R): Intense summer heating of the continental landmass creates a thermal low over Northwest India, drawing moisture-laden winds from the Indian Ocean.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.12

Assertion (A): The Tibetan Plateau acts as a thermal heat engine that reinforces the South Asian summer monsoon circulation.

Reason (R): Intense summer insolation on the elevated Tibetan plateau produces a strong convective thermal low, pumping air aloft and generating the Tropical Easterly Jet stream.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.13

Assertion (A): Tamil Nadu's Coromandel Coast receives the majority of its annual rainfall during the winter months (October to December).

Reason (R): The Northeast retreating monsoon winds absorb moisture while crossing the warm waters of the Bay of Bengal before hitting the Coromandel coast.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.14

Assertion (A): Western Rajasthan receives negligible rainfall from the Arabian Sea branch of the Southwest Monsoon.

Reason (R): The Aravalli Range lies parallel to the direction of the Arabian Sea monsoon winds, providing no orographic barrier to induce precipitation.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): Tropical cyclones develop readily over cold polar ocean waters below 10°C.

Reason (R): Tropical cyclogenesis requires a sea surface temperature of at least 26.5°C to 27°C to sustain vigorous evaporation and latent heat release.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): The eye of a mature tropical cyclone is a region of violent storm winds and torrential downpours.

Reason (R): The cyclone eye experiences subsiding dry air, light winds, and clear or partly cloudy skies with the lowest central surface pressure.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): Mawsynram in Meghalaya receives the highest average annual rainfall in the world.

Reason (R): The funnel-shaped Khasi Hills trap moisture-bearing Bay of Bengal monsoon winds, forcing them to rise abruptly and undergo intense orographic condensation.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): Pre-monsoon thunder squalls in West Bengal are locally referred to as 'Kalbaisakhi'.

Reason (R): Kalbaisakhi storms originate from the clash of hot, dry continental air with moist maritime Bay of Bengal air, producing severe squalls and hailstorms.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & INSTITUTIONAL COMPLEXITIES (Q.19 TO Q.25) [7 MARKS]**Q.19**

Assertion (A): An El Niño event in the equatorial Pacific Ocean is often correlated with weak or deficit monsoon rainfall in India.

Reason (R): El Niño warms the eastern Pacific, displacing the atmospheric Walker circulation and weakening low-pressure convective cells over the Indian subcontinent.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): The ozone hole over Antarctica is primarily caused by elevated emissions of carbon dioxide (\$CO_2\$).

Reason (R): Chlorofluorocarbons (CFCs) release reactive chlorine radicals under polar stratospheric cloud chemistry that catalytically destroy ozone (\$O_3\$) molecules.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): Melting of sea ice in the Arctic ocean triggers a positive feedback loop that accelerates global warming.

Reason (R): White reflective sea ice with high albedo is replaced by dark ocean water with low albedo, causing the ocean to absorb more solar radiation.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): Places located near maritime coasts experience a much lower diurnal and annual range of temperature than inland continental locations.

Reason (R): Water has a significantly higher specific heat capacity than land, causing oceans to heat up and cool down much slower than continental land surfaces.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): Global warming causes thermal expansion of ocean water, contributing to global sea-level rise.

Reason (R): Liquid water expands in volume as its temperature rises above 4°C, increasing the volumetric height of global oceanic basins.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): The 'Loo' winds of North India in May and June are cool, humid maritime sea breezes.

Reason (R): The Loo consists of strong, extremely hot, and dry westerly winds blowing from the arid Thar desert across the northern plains.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): A high carbon footprint indicates that an individual or activity contributes substantially to anthropogenic greenhouse gas emissions.

Reason (R): The carbon footprint quantifies total direct and indirect emissions of greenhouse gases, expressed in equivalents of carbon dioxide (\$CO_2e\$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01 (A) (B) (C) (D)
06 (A) (B) (C) (D)
11 (A) (B) (C) (D)
16 (A) (B) (C) (D)
21 (A) (B) (C) (D)

02 (A) (B) (C) (D)
07 (A) (B) (C) (D)
12 (A) (B) (C) (D)
17 (A) (B) (C) (D)
22 (A) (B) (C) (D)

03 (A) (B) (C) (D)
08 (A) (B) (C) (D)
13 (A) (B) (C) (D)
18 (A) (B) (C) (D)
23 (A) (B) (C) (D)

04 (A) (B) (C) (D)
09 (A) (B) (C) (D)
14 (A) (B) (C) (D)
19 (A) (B) (C) (D)
24 (A) (B) (C) (D)

05 (A) (B) (C) (D)
10 (A) (B) (C) (D)
15 (A) (B) (C) (D)
20 (A) (B) (C) (D)
25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & HISTORICAL/GEOGRAPHICAL CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Causal Rationale & Conceptual Explanation
Q.1	(A)	Assertion is true. Reason is true and explains why meteorological precipitation cycles require the dense, moisture-rich troposphere.
Q.2	(A)	Assertion is true. Reason is true and provides the meteorological reason for high-altitude aeronautical cruising.
Q.3	(D)	Assertion is false: Temperature increases in the stratosphere and thermosphere. Reason is true: Stratospheric ozone absorption creates a thermal inversion.
Q.4	(A)	Assertion is true. Reason is true and explains why proximity to the warm terrestrial surface dictates tropospheric temperature gradients.
Q.5	(A)	Assertion is true. Reason is true and defines Ferrel's Law and its planetary rotational origin.
Q.6	(D)	Assertion is false: Coriolis force is zero at the equator ($\sin 0^\circ = 0$) and maximum at the poles ($\sin 90^\circ = 1$). Reason is true: The mathematical formula contains $\sin \phi$.
Q.7	(A)	Assertion is true. Reason is true and explains thermal low-pressure generation at the ITCZ.
Q.8	(D)	Assertion is false: Subtropical highs are dynamically produced by subsidence, not cold ground. Reason is true: Hadley cell downward subsidence creates dynamic high pressure.
Q.9	(A)	Assertion is true. Reason is true and explains the biological necessity of natural thermal blanket trapping of terrestrial infrared rays.
Q.10	(D)	Assertion is false: Primary greenhouse gases are polyatomic (H_2O , CO_2 , CH_4). Reason is true: Homonuclear diatomics do not possess infrared absorption dipoles.
Q.11	(A)	Assertion is true. Reason is true and explains the differential thermal land-sea heating mechanism driving the Southwest Monsoon.
Q.12	(A)	Assertion is true. Reason is true and details the role of Tibetan high-altitude sensible heating in maintaining monsoon jet streams.
Q.13	(A)	Assertion is true. Reason is true and explains why Tamil Nadu is rain-shadowed during SW monsoon but wet during NE retreating monsoon.
Q.14	(A)	Assertion is true. Reason is true and explains the orographic alignment responsible for the aridity of western Rajasthan.
Q.15	(D)	Assertion is false: Cyclones require warm tropical seas. Reason is true: High thermal energy and latent heat from warm waters are mandatory for cyclonic development.
Q.16	(D)	Assertion is false: The eye is calm and rainless, while the eyewall has the violent storms. Reason is true: Central subsidence suppresses clouds in the eye.
Q.17	(A)	Assertion is true. Reason is true and explains the topographic funneling mechanism causing exceptional precipitation at Mawsynram.
Q.18	(A)	Assertion is true. Reason is true and explains the thermodynamic instability generating Kalbaisakhi (Nor'westers).
Q.19	(A)	Assertion is true. Reason is true and explains how ENSO-induced Walker cell dislocation suppresses Indian monsoon convection.
Q.20	(D)	Assertion is false: Ozone depletion is caused by CFCs and halons, not CO_2 . Reason is true: Catalytic chlorine cycles on PSCs break down stratospheric ozone.

Q. No.	Correct	Causal Rationale & Conceptual Explanation
Q.21	(A)	Assertion is true. Reason is true and explains the ice-albedo positive feedback mechanism.
Q.22	(A)	Assertion is true. Reason is true and provides the thermodynamic explanation for maritime moderate climates vs continental extremes.
Q.23	(A)	Assertion is true. Reason is true and explains why steric thermal expansion is a primary driver of rising sea levels alongside glacial melt.
Q.24	(D)	Assertion is false: Loo is scorching hot and dry, not cool or maritime. Reason is true: The Loo blows out of desert interiors across North India.
Q.25	(A)	Assertion is true. Reason is true and defines the environmental metric used to assess climate impact.